3.4 Factorisation using identities

A factor of a number is the number that exactly divides a given number without leaving any remainder. For example, the factors of 4 are 1, 2 and 4; the factors of 12 are 1, 2, 3, 4, 6 and 12. And, the number can be expressed as product of its factors such as $12 = 1 \times 12$ or 2×6 or 3×4.

In the same way, algebraic expressions also have factors that divide the expression exactly. Given an algebraic expression, the **factors of that algebraic expression** are two or more expressions whose product is the given expression. For example, $x^2y = x \times x \times y$, hence the factors of the algebraic expression x^2y are x, x and y.

Consider the algebraic expression $x^2 - 4$.

Now, $x^2 - 4 = x^2 - 2^2 = (x + 2)(x - 2)$ [because $a^2 - b^2 = (a+b)(a-b)$]

Therefore, $(x + 2)$ and $(x - 2)$ are the factors of $x^2 - 4$.

Obviously, 1 is also a factor of $x^2 - 4$.

We know that, $x^2 = x \times x$. Similarly, $(x+5)^2 = (x+5) \times (x+5)$.

Thus, the factors of $(x + 5)^2$ are $(x + 5)$ and $(x + 5)$.

The process of writing an algebraic expression as the product of its factors is called factorisation.

We shall learn more to factorise algebraic expressions, using the basic identities.

Example 3.5 Express the following algebraic expressions as the product of its factors:
(i) ab^2 (ii) $-3pq^3$ (iii) $12m^2n^2p$

Solution

(i) $ab^2 = a \times b \times b$.

(ii) $-3pq^3 = -3 \times p \times q \times q \times q$.

(iii) $12\,m^2n^2p = 4 \times 3 \times m \times m \times n \times n \times p = 2 \times 2 \times 3 \times m \times m \times n \times n \times p$.

Example 3.6 Factorise by using the identity: $a^2 - b^2 = (a + b)(a - b)$
(i) $a^2 - 1$ (ii) $9k^2 - 25$ (iii) $64x^2 - 81y^2$

Solution

(i) $a^2 - 1 = a^2 - 1^2 = (a + 1)(a - 1)$ [since, $1^2 = 1 \times 1 = 1$]
Therefore, the factors of $a^2 - 1$ are $(a + 1)$ and $(a - 1)$.

(ii) $9k^2 - 25 = \left(3^2 \times k^2\right) - 5^2 = \left(3k\right)^2 - 5^2$ [since, $a^m \times b^m = (a \times b)^m$]
$= (3k + 5)(3k - 5)$ $\left[\text{since, } a^2 - b^2 = (a+b)(a-b)\right]$

Therefore, the factors of $9k^2 - 25$ are $(3k + 5)$ and $(3k - 5)$.

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